Information Retrieval with BM25 and Embeddings

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0. Imports and helper functions

In [1]:
```
import gzip
import json
import pickle
import re
import string
import pandas as pd
from IPython.display import display, Markdown
from textwrap import shorten

# Get the root
import sys
import os
project_root = os.path.abspath(os.path.join(os.getcwd(), ".."))
sys.path.append(project_root)

# Suppress warnings
import warnings
warnings.filterwarnings('ignore')

#import nltk
#nltk.download("stopwords")
from nltk.corpus import stopwords
from rank_bm25 import BM25Okapi
from langchain_core.documents import Document
from sentence_transformers import SentenceTransformer

# Import scripts
from src.bm25 import build_bm25, save_bm25, load_bm25, bm25_search
from src.semantic import (
    build_embeddings,
    build_faiss_index,
    save_faiss,
    load_faiss,
    semantic_search
)
```

In [2]:
```
def load_jsonl_gz(path, n=5000):
    '''
    load first N records from a .jsonl.gz file
    '''
    records = []
    with gzip.open(path, "rt", encoding="utf-8") as f:
```

```
        for i, line in enumerate(f):
            if i >= n:
                break
            try:
                records.append(json.loads(line))
            except json.JSONDecodeError:
                continue
    return records
```

In [3]:

```
def display_search_results(results, query, system_name, max_chars=200):
    '''
    Pretty display for retrieval results.
    '''
    rows = []

    for rank, r in enumerate(results, start=1):
        metadata = r["metadata"]
        source = metadata["source"]

        raw_text = r["text"].strip()
        parts = [p.strip() for p in raw_text.split("\n\n") if p.strip()]

        title = parts[0]
        body = " ".join(parts[1:]) if len(parts) > 1 else raw_text
        body = " ".join(body.split())

        if source == "metadata":
            categories = metadata.get("categories", [])
            categories_str = " > ".join(categories[:4]) if isinstance(
                categories, list) else ""
            info = metadata.get("main_category", "")
            if categories_str:
                info = f"{info} | {categories_str}" if info else categories_
        else:
            info = f'rating={metadata.get("rating", "")}, verified={metadata

        rows.append({
            "Rank": rank,
            "Score": float(r["score"]),
            "Source": source,
            "ASIN": metadata["asin"],
            "Title": title,
            "Info": info,
            "Snippet": shorten(body, width=max_chars, placeholder="...")
        })

    df = pd.DataFrame(rows)

    title = f"#### {system_name} results for: '{query}'"
    display(Markdown(title))

    display(
        df.style
        .hide(axis="index")
        .format({"Score": "{:.3f}"})
        .set_properties(subset=["Title", "Info", "Snippet"], **{"text-align"
```

```
        .set_properties(subset=["Snippet"], **{"white-space": "pre-wrap"})
    )

    return df
```

1. Data exploration and preprocessing

In [4]:
```
# Load small subsets (5000 rows)
meta_path = "../data/raw/meta_Appliances.jsonl.gz"
review_path = "../data/raw/Appliances.jsonl.gz"

meta_records = load_jsonl_gz(meta_path, n=5000)
review_records = load_jsonl_gz(review_path, n=5000)

print(f"Loaded {len(meta_records)} metadata records")
print(f"Loaded {len(review_records)} review records")

# Convert to DataFrames for easier EDA
meta_df = pd.DataFrame(meta_records)
review_df = pd.DataFrame(review_records)

meta_df.head()
```

```
Loaded 5000 metadata records
Loaded 5000 review records
```

Out[4]:

	main_category	title	average_rating	rating_number	features	
0	Industrial & Scientific	ROVSUN Ice Maker Machine Countertop, Make 44lb...	3.7	61	[【Quick Ice Making】This countertop ice machine...	
1	Tools & Home Improvement	HANSGO Egg Holder for Refrigerator, Deviled Eg...	4.2	75	[Plastic, Practical Kitchen Storage - Our egg ...	
2	Tools & Home Improvement	Clothes Dryer Drum Slide, General Electric, Ho...	3.5	18	[]	[Brar drum sl
3	Tools & Home Improvement	154567702 Dishwasher Lower Wash Arm Assembly f...	4.5	26	[MODEL NUMBER:154567702 Dishwasher Lower Wash ...	NUMBEI Dishw
4	Tools & Home Improvement	Whirlpool W10918546 Igniter	3.8	12	[This is a Genuine OEM Replacement Part.]	[Whirl

1.1. Dataset overview

In [5]:

```
print("Metadata fields:", meta_df.columns.tolist())
print("Review fields:", review_df.columns.tolist())
print()
print("Metadata shape:", meta_df.shape)
print("Review shape:", review_df.shape)
```

```
Metadata fields: ['main_category', 'title', 'average_rating', 'rating_numbe
r', 'features', 'description', 'price', 'images', 'videos', 'store', 'catego
ries', 'details', 'parent_asin', 'bought_together']
Review fields: ['rating', 'title', 'text', 'images', 'asin', 'parent_asin',
'user_id', 'timestamp', 'helpful_vote', 'verified_purchase']

Metadata shape: (5000, 14)
Review shape: (5000, 10)
```

1.2. Sample records

In [6]:

```
meta_df.sample(3)
review_df.sample(3)
```

Out[6]:

	rating	title	text	images	asin	parent_asin	
2762	5.0	Nice Finish For Range Top	The Corelle Coordinates Burner Covers match my...	[]	B000UOG7CG	B000UOG7CG	AH5UPJBD6NF
4303	4.0	Delivered as ordered.	Works as expected.	[]	B07S7WF3S3	B0C64H22TB	AFNG4ICJNWPS
2835	5.0	Convenient	Convenient	[]	B0728JN794	B09W8W65XY	AHDLTVY2XJMU

1.3. Data preprocessing

In [7]:

```
# Fields we decided to use for retrieval
META_TEXT_FIELDS = ["title", "description", "features"]
REVIEW_TEXT_FIELDS = ["title", "text"]

documents = []

# Process metadata entries
for _, row in meta_df.iterrows():
    text_parts = []

    for field in META_TEXT_FIELDS:
        value = row.get(field)
        if isinstance(value, str) and len(value.strip()) > 0:
            text_parts.append(value.strip())

    # Skip if no meaningful text
    if len(text_parts) == 0:
        continue

    full_text = "\n\n".join(text_parts)

    documents.append(
        Document(
            page_content=full_text,
            metadata={
                "source": "metadata",
                "asin": row.get("parent_asin"),
                "main_category": row.get("main_category"),
                "categories": row.get("categories"),
            }
        )
    )

# Process review entries
for _, row in review_df.iterrows():
    text_parts = []

    for field in REVIEW_TEXT_FIELDS:
        value = row.get(field)
```

```
        if isinstance(value, str) and len(value.strip()) > 0:
            text_parts.append(value.strip())

    if len(text_parts) == 0:
        continue

    full_text = "\n\n".join(text_parts)

    documents.append(
        Document(
            page_content=full_text,
            metadata={
                "source": "review",
                "asin": row.get("asin"),
                "rating": row.get("rating"),
                "verified_purchase": row.get("verified_purchase"),
            }
        )
    )

len(documents)
```

Out[7]: 9999

1.3.1 Selection and justification of fields for retrieval

For retrieval, we focused on fields that contain meaningful natural-language text.

- From the metadata, `title`, `description`, and sometimes `features` provide the most useful product information.
- From the reviews, the key fields are `text` (full review body) and `title` (short summary).
  Other fields like price, images, or ratings don't add much value for text-based retrieval.

1.3.2. Description of text preprocessing

Light preprocessing was applied to keep the text clean but still semantically rich:

- all text lowercased
- extra whitespace and URLs removed
- punctuation kept
- empty or extremely short entries removed

There is no stemming or lemmatization involved, as modern embedding models handle them well on their own.

2. Keyword-based retrieval system (BM25)

In [8]:
```
bm25, tokenized = build_bm25(documents)
save_bm25(bm25, tokenized, save_dir="../bm25_index/")
```

```
bm25, tokenized = load_bm25("../bm25_index/")
```

2.1. Example query results

In [9]:
```
query = "quiet dishwasher stainless steel"
results = bm25_search(query, bm25, documents, k=5)

display_search_results(results, query=query, system_name="BM25 search");
```

BM25 search results for: 'quiet dishwasher stainless steel'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	12.852	metadata	B078T6SMBM	Dishwasher Covers Brushed Stainless Steel Panels	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories	Dishwasher Covers Brushed Stainless Steel Panels
2	12.443	metadata	B004TSON5M	EASTMAN 8 Stainless Steel Dishwasher Hose 98524	Appliances | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Hoses	EASTMAN 8 Stainless Steel Dishwasher Hose 98524
3	12.443	metadata	B00NN1CAJW	Front Control Dishwasher in White with Stainless Steel Tub	Tools & Home Improvement | Appliances > Dishwashers > Built-In Dishwashers	Front Control Dishwasher in White with Stainless Steel Tub
4	11.963	review	B013CBE2DE	Relatively easy. The dishwasher is nice and quiet again	rating=5.0, verified=True	Used a youtube video for how to install. Relatively easy. The dishwasher is nice and quiet again.
5	11.699	metadata	B0C2K57SHK	EASTMAN 41042 Stainless Steel Dishwasher Connector, 5 Ft, Silver	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Hoses	EASTMAN 41042 Stainless Steel Dishwasher Connector, 5 Ft, Silver

3. Semantic-based retrival system

3.1. Embeddings with sentence transformers

In [10]:
```
embeddings, model = build_embeddings(documents)
index = build_faiss_index(embeddings)
save_faiss(index, documents, save_dir="../semantic_index/")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set
a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights:   0%|          | 0/103 [00:00<?, ?it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architectur
e; not ok if you expect identical arch.
Batches:   0%|          | 0/313 [00:00<?, ?it/s]
```

3.2. FAISS index and documents

In [11]:
```
index, documents = load_faiss("../semantic_index/")
```

3.3. Example query results

In [12]:
```
query = "quiet dishwasher stainless steel"
results = semantic_search(query, index, model, documents, k=5)

display_search_results(results, query=query, system_name="Semantic search");
```

Semantic search results for: 'quiet dishwasher stainless steel'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.719	review	B0074CK3UO	poor performer	rating=2.0, verified=True	as some others have indicated, this dishwasher seems to do very little, if anything, in the way of actually cleaning dishes. :(i don't mind the noise level a bit; i guess i grew up with noisier...
2	0.722	metadata	B078T6SMBM	Dishwasher Covers Brushed Stainless Steel Panels	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories	Dishwasher Covers Brushed Stainless Steel Panels
3	0.726	metadata	B00NN1CAJW	Front Control Dishwasher in White with Stainless Steel Tub	Tools & Home Improvement | Appliances > Dishwashers > Built-In Dishwashers	Front Control Dishwasher in White with Stainless Steel Tub
4	0.744	review	B07664NDKB	My dishwasher is finally quiet	rating=5.0, verified=True	Doing dishes had been a noisy ordeal for a long time. The banging noise was awful. Finally, thanks to Google, I figured out what was wrong and identified this part as the cause and Amazon had it...
5	0.767	metadata	B01G9ASTG6	GE CDT835SSJSS Cafe 24" Stainless Steel Fully Integrated Dishwasher - Energy Star	Appliances | Appliances > Dishwashers > Built-In Dishwashers	GE CDT835SSJSS Cafe 24" Stainless Steel Fully Integrated Dishwasher - Energy Star

4. Qualitative evaluation

In [13]:
```
queries = [
    "stainless steel dishwasher",
    "refrigerator water filter",
```

```
    "gas range 30 inch",
    "something to keep drinks cold in a dorm room",
    "appliance for washing dishes quietly at night",
    "replacement part to stop a dishwasher from leaking at the bottom",
    "small machine for drying clothes in an apartment",
    "best dishwasher for a small apartment under $800",
    "energy-efficient refrigerator for a family of four",
    "reliable stove for frequent home cooking that is easy to clean",
]
```

In [14]:

```
k = 5

for query in queries:
    bm25_results = bm25_search(query, bm25, documents, k=k)
    semantic_results = semantic_search(query, index, model, documents, k=k)

    bm25_df = display_search_results(
        bm25_results, query=query, system_name="BM25")

    semantic_df = display_search_results(
        semantic_results, query=query, system_name="Semantic")
```

BM25 results for: 'stainless steel dishwasher'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	12.852	metadata	B078T6SMBM	Dishwasher Covers Brushed Stainless Steel Panels	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories	Dishwasher Covers Brushed Stainless Steel Panels
2	12.443	metadata	B00NN1CAJW	Front Control Dishwasher in White with Stainless Steel Tub	Tools & Home Improvement | Appliances > Dishwashers > Built-In Dishwashers	Front Control Dishwasher in White with Stainless Steel Tub
3	12.443	metadata	B004TSON5M	EASTMAN 8 Stainless Steel Dishwasher Hose 98524	Appliances | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Hoses	EASTMAN 8 Stainless Steel Dishwasher Hose 98524
4	11.699	metadata	B0052EK0MW	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel	Appliances | Appliances > Dishwashers > Built-In Dishwashers	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel
5	11.699	metadata	B0C2K57SHK	EASTMAN 41042 Stainless Steel Dishwasher Connector, 5 Ft, Silver	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Hoses	EASTMAN 41042 Stainless Steel Dishwasher Connector, 5 Ft, Silver

Semantic results for: 'stainless steel dishwasher'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.536	metadata	B00NN1CAJW	Front Control Dishwasher in White with Stainless Steel Tub	Tools & Home Improvement | Appliances > Dishwashers > Built-In Dishwashers	Front Control Dishwasher in White with Stainless Steel Tub
2	0.563	metadata	B078T6SMBM	Dishwasher Covers Brushed Stainless Steel Panels	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories	Dishwasher Covers Brushed Stainless Steel Panels
3	0.604	review	B08H4KYL4T	Dishwasher	rating=5.0, verified=True	Nice to have, Thank You
4	0.623	metadata	B0052EK0MW	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel	Appliances | Appliances > Dishwashers > Built-In Dishwashers	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel
5	0.633	metadata	B01G9ASTG6	GE CDT835SSJSS Cafe 24" Stainless Steel Fully Integrated Dishwasher - Energy Star	Appliances | Appliances > Dishwashers > Built-In Dishwashers	GE CDT835SSJSS Cafe 24" Stainless Steel Fully Integrated Dishwasher - Energy Star

BM25 results for: 'refrigerator water filter'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	10.487	metadata	B08NYLG5VD	DA97-17376B Refrigerator Water Filter, Samsung 1 pack Refrigerator Water Filter	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	DA97-17376B Refrigerator Water Filter, Samsung 1 pack Refrigerator Water Filter
2	10.100	review	B0169T71UW	Great refrigerator filter	rating=5.0, verified=True	I am very happy with the water filter for my refrigerator. The water already tastes so much better.
3	9.399	review	B012V6DYT4	Refrigerator Water Filter Replacement	rating=5.0, verified=True	Perfect replacement for my Samsung water filter. Works great.
4	9.300	metadata	B09P3CNG65	Refrigerator wat/er fil/ter F1-3 pcs-YUGg-004	Tools & Home Improvement | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	Refrigerator wat/er fil/ter F1-3 pcs-YUGg-004
5	9.201	metadata	B00YD328X2	Replacement for MZD2665HEB Refrigerator Water Filter - Compatible with Maytag UKF8001 Fridge Water Filter Cartridge	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	Replacement for MZD2665HEB Refrigerator Water Filter - Compatible with Maytag UKF8001 Fridge Water Filter Cartridge

Semantic results for: 'refrigerator water filter'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.331	review	B00UB38V2A	Fridge water filter	rating=5.0, verified=True	I’ve used this ever since I bought my fridge. No complaints
2	0.348	review	B009PCI2JU	GE refrigerator water filter	rating=4.0, verified=True	Works well. Very expensive!
3	0.418	metadata	B07Y57RPX4	water filter	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	water filter
4	0.418	metadata	B07Y58LW1P	water filter	Appliances | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	water filter
5	0.418	metadata	B08CK3PXFZ	water filter	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	water filter

BM25 results for: 'gas range 30 inch'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	16.297	metadata	B08XZ75LK4	Empava 30" Slide-in Single Oven Gas Range with 4 Sealed Ultra High-Low Burners-Heavy Duty Continuous Grates in Stainless Steel, 30 Inch	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Slide-In Ranges	Empava 30" Slide-in Single Oven Gas Range with 4 Sealed Ultra High-Low Burners-Heavy Duty Continuous Grates in Stainless Steel, 30 Inch
2	15.966	metadata	B08TX564PQ	Empava 30" Slide-in Single Oven Gas Range with 4 Sealed Ultra High-Low Burners-Heavy Duty Continuous Grates in Stainless Steel, 30 Inch, Silver	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	Empava 30" Slide-in Single Oven Gas Range with 4 Sealed Ultra High-Low Burners-Heavy Duty Continuous Grates in Stainless Steel, 30 Inch, Silver
3	15.387	metadata	B01AFU5OW8	30 inch Gas Cooktop Gas Cooktops 5 Burner Built-In Gas Stoves NG LPG Gas Stove Range with 5 High Efficiency Burners NG/LPG Convertible Stainless Steel Gas Stove Top with Thermocouple Protection	Appliances	30 inch Gas Cooktop Gas Cooktops 5 Burner Built-In Gas Stoves NG LPG Gas Stove Range with 5 High Efficiency Burners NG/LPG Convertible Stainless Steel Gas Stove Top with Thermocouple Protection
4	15.334	metadata	B01HJOY8J8	THOR KITCHEN HRG3026U 30 inch Gas Range with 4.2 cu. ft. Oven, 4 Burners, Convection Fan, Stainless Steel	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	THOR KITCHEN HRG3026U 30 inch Gas Range with 4.2 cu. ft. Oven, 4 Burners, Convection Fan, Stainless Steel
5	14.973	metadata	B07S1WKG9Q	Thor Kitchen 30 inch Freestanding Pro-Style Gas Range with 4.55 cu.ft. Oven, 5 Burners, in Stainless Steel - LRG3001U + LP Kit	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	Thor Kitchen 30 inch Freestanding Pro-Style Gas Range with 4.55 cu.ft. Oven, 5 Burners, in Stainless Steel - LRG3001U + LP Kit

Semantic results for: 'gas range 30 inch'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.598	metadata	B07S1WKG9Q	Thor Kitchen 30 inch Freestanding Pro-Style Gas Range with 4.55 cu.ft. Oven, 5 Burners, in Stainless Steel - LRG3001U + LP Kit	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	Thor Kitchen 30 inch Freestanding Pro-Style Gas Range with 4.55 cu.ft. Oven, 5 Burners, in Stainless Steel - LRG3001U + LP Kit
2	0.608	metadata	B002LK46L6	NXR DRGB3602 Professional Style Gas Range, 36", Stainless Steel	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	NXR DRGB3602 Professional Style Gas Range, 36", Stainless Steel
3	0.656	metadata	B01M32MY26	Blomberg BGRP34520SS 30" Pro Gas Range, Stainless Steel	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	Blomberg BGRP34520SS 30" Pro Gas Range, Stainless Steel
4	0.711	metadata	B078L93FYC	NXR DRGB3602BD 36" Professional Style Gas Range with Range Hood, Stainless Steel	Appliances | Appliances > Ranges, Ovens & Cooktops > Range Hoods	NXR DRGB3602BD 36" Professional Style Gas Range with Range Hood, Stainless Steel
5	0.719	metadata	B001MBCPK4	X304GGVVI Bertazzoni 30" Pro Gas Range with 4 Burners - Burgundy	Amazon Home | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	X304GGVVI Bertazzoni 30" Pro Gas Range with 4 Burners - Burgundy

BM25 results for: 'something to keep drinks cold in a dorm room'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	13.492	review	B08ZYJ8CRX	Love to use a lot everyday!!	rating=5.0, verified=True	Family love it and use for cold drinks !!
2	10.940	review	B09ZKBRNPC	Sturdy reusable hot or cold cups	rating=5.0, verified=False	I ordered these for our picnic kit. They are sturdy plastic, textured on the outside and smooth on the inside, great for either iced or hot drinks. I ran them through the dishwasher and they were...
3	9.176	metadata	B09X4L5N4B	Portable AC,Portable Air Conditioner Fan Personal Cooler Fan Table Humidifier Rotating USB Power Supply Timer LED Light 4 Speeds for Bedroom Office Living Room Dorm (white)	Appliances	Portable AC,Portable Air Conditioner Fan Personal Cooler Fan Table Humidifier Rotating USB Power Supply Timer LED Light 4 Speeds for Bedroom Office Living Room Dorm (white)
4	8.995	metadata	B0BTBPW4KB	Portable Ice Maker for Countertop, 9 Ice Cube Ready in 7 Mins, 26 lbs in 24 hrs, Ice Maker Machine with Basket and Scoop for Home Bar Office Dorm Camping RV Parties Drinks, 2 Size Bullet Ice	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Ice Makers	Portable Ice Maker for Countertop, 9 Ice Cube Ready in 7 Mins, 26 lbs in 24 hrs, Ice Maker Machine with Basket and Scoop for Home Bar Office Dorm Camping RV Parties Drinks, 2 Size Bullet Ice
5	8.965	review	B098HD9PKL	Great for dorm Rooms and more...	rating=5.0, verified=False	This fits under my bed in the dorm room perfectly. I really like the reversible door. That is so handy for anytime I rearrange or my roommate wants to rearrange the

Rank	Score	Source	ASIN	Title	Info	Snippet
						room. Cans fit on the shelves...

Semantic results for: 'something to keep drinks cold in a dorm room'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.855	review	B098H39K3W	Perfect for Small Spaces	rating=4.0, verified=False	This is the perfect little fridge for small spaces. Ideal for a dorm, workshop, bar or garage, it keeps things cool. The little freezer on top can make ice cubes. It is quiet and has performed...
2	0.926	metadata	B000XYN5RS	Mini USB-Powered Fridge Cooler for Beverage Drink Cans in Cubicle and Home office (Red)	All Electronics | Appliances > Refrigerators, Freezers & Ice Makers > Beverage Refrigerators	Mini USB-Powered Fridge Cooler for Beverage Drink Cans in Cubicle and Home office (Red)
3	0.933	metadata	B005JAVD3Y	IMAGE® Mini USB-Powered Fridge Cooler for Beverage Drink Cans in Cubicle and Home office (Black)	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Beverage Refrigerators	IMAGE® Mini USB-Powered Fridge Cooler for Beverage Drink Cans in Cubicle and Home office (Black)
4	0.959	review	B098HD9PKL	Great for dorm Rooms and more…	rating=5.0, verified=False	This fits under my bed in the dorm room perfectly. I really like the reversible door. That is so handy for anytime I rearrange or my roommate wants to rearrange the room. Cans fit on the shelves...
5	0.974	metadata	B0895L6PLD	COOLLIFE Compact Refrigerator Freezer, Retro Style Single Door Mini Fridge, 1.6 Cu.Ft Home Small Fridge Cooler with Two Glass Shelf, Using for Dorm, Office, Dorm	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Beverage Refrigerators	COOLLIFE Compact Refrigerator Freezer, Retro Style Single Door Mini Fridge, 1.6 Cu.Ft Home Small Fridge Cooler with Two Glass Shelf, Using for Dorm, Office, Dorm

BM25 results for: 'appliance for washing dishes quietly at night'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	9.553	review	B004Q3WUI2	NICE TO HAVE CLEAN DISHES AGAIN	rating=5.0, verified=True	Nice to see sparking clean dishes and glasses again!
2	8.862	review	B075TBB334	Perfect solution!	rating=5.0, verified=True	My dishwasher has no indicator light when dishes are clean so we ended up mixing in more dirty dishes. This is a perfect solution
3	8.507	metadata	B01N19MVJB	ATB White Vinyl Quilted Washing Machine Dryer Cover Top Load Dust Zippered Appliance	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	ATB White Vinyl Quilted Washing Machine Dryer Cover Top Load Dust Zippered Appliance
4	8.405	review	B078MMXJPC	Visible difference	rating=5.0, verified=True	Has saved our dishes! We don’t over wash clean dishes and we keep track of what’s dirty or not more easily. Like the big print.
5	8.230	metadata	B000UXDQ8K	Ibis & Orchid Camelia Night Light #50062	Tools & Home Improvement | Appliances > Parts & Accessories	Ibis & Orchid Camelia Night Light #50062

Semantic results for: 'appliance for washing dishes quietly at night'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.781	review	B01I6MHNQC	Amazing small dishwasher - so quiet!	rating=5.0, verified=True	We use this as a second dishwasher for our cat dishes. It is quiet and gets the dishes really clean. We are a Danby repeat customer and I can highly recommend their dishwashers. This little...
2	0.898	review	B004N4MSPO	Nice addition to my little kitchen.	rating=4.0, verified=True	Easy set up, fairly quiet. So happy to find a dishwasher that works in my tiny 50’s kitchen. Didn’t exactly fit under the cabinet so I got a rolling stand. Doesn’t vibrate much. Very stable on the...
3	0.905	review	B096RYWSQM	Bringing it clean	rating=5.0, verified=True	This is a good machine. It gets the dishes clean, it is quiet, and efficient. Great value!
4	0.940	review	B00PI74MMM	SO NOISY!!!	rating=1.0, verified=False	oh my gosh, this is the noisiest machine I have ever owned! you can hear the squealing and knocking across the house. just can't believe such an expensive, name brand washer sounds like it does. on...
5	0.948	review	B07664NDKB	My dishwasher is finally quiet	rating=5.0, verified=True	Doing dishes had been a noisy ordeal for a long time. The banging noise was awful. Finally, thanks to Google, I figured out what was wrong and identified this part as the cause and Amazon had it...

BM25 results for: 'replacement part to stop a dishwasher from leaking at the bottom'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	13.458	metadata	B07BBK55DP	WD12X10304 Slide Stop Replacement for GE Dishwasher Replace AP4484666 WD12X344	Tools & Home Improvement | Appliances > Parts & Accessories	WD12X10304 Slide Stop Replacement for GE Dishwasher Replace AP4484666 WD12X344
2	12.351	metadata	B07WXHFQ68	Supplying Demand W10195622 Dishwasher Upper Dishrack Slide Rail Rear Stop Clip Replacement	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Baskets	Supplying Demand W10195622 Dishwasher Upper Dishrack Slide Rail Rear Stop Clip Replacement
3	11.273	metadata	B009IO1C92	Dishwasher Upper Rack Stop for General Electric, Hotpoint, WD12X10304	Tools & Home Improvement | Appliances > Parts & Accessories > Dishwasher Parts & Accessories	Dishwasher Upper Rack Stop for General Electric, Hotpoint, WD12X10304
4	11.264	review	B08FGKR6P1	Replacement part.	rating=5.0, verified=True	Fixed my broken dishwasher.
5	10.735	review	B08QTQW9VQ	It leaking on bottom but those company is out of business so can’t replace!!	rating=3.0, verified=True	Love it but can’t replace because of company is out of business so I brought different one and good so far

Semantic results for: 'replacement part to stop a dishwasher from leaking at the bottom'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.502	review	B08FGKR6P1	Replacement part.	rating=5.0, verified=True	Fixed my broken dishwasher.
2	0.609	review	B00CDS7ZUU	Perfect Part	rating=5.0, verified=True	This part saved me a ton of money on dishwasher repair or replacement. It was reasonable to install though not easy. (Not the fault of this part... it just is what it is.) Works great. No more leaks!
3	0.651	review	B00HNEJ0V6	simple an quick	rating=5.0, verified=False	Simple to install replacement part for your dishwasher, even if you never worked on a dishwasher before.. This parts located in the front left hand side of the washer underneath.. Tends to be your...
4	0.743	review	B08QTQW9VQ	It leaking on bottom but those company is out of business so can’t replace!!	rating=3.0, verified=True	Love it but can’t replace because of company is out of business so I brought different one and good so far
5	0.804	review	B07DKFKJF1	Excellent Replacement Part, Better Than The Original!	rating=5.0, verified=True	My rack wheels kept breaking, the original wheel assembly is not made very well. These replacement wheel assemblies are much better, stronger and will hold up much longer. If this fits your...

BM25 results for: 'small machine for drying clothes in an apartment'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	16.269	metadata	B09N8NXR35	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine
2	15.330	metadata	B0C5RRJ9C6	Portable Washing Machine,Foldable Mini Small Washing Machine for Underwear, Socks, Baby clothes, Apartment, Laundry, Camping, RV, Travel (110V-240V)-Best Gift for Friend(pink)	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	Portable Washing Machine,Foldable Mini Small Washing Machine for Underwear, Socks, Baby clothes, Apartment, Laundry, Camping, RV, Travel (110V-240V)-Best Gift for Friend(pink)
3	14.037	metadata	B08KQ2T7HF	Portable Stacked Dryer, Multifunctional Mini Manual Dryer Machine Hand-Operated Dryer Spinner, Non-Electric Compact Laundry Machine for Drying Clothes	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	Portable Stacked Dryer, Multifunctional Mini Manual Dryer Machine Hand-Operated Dryer Spinner, Non-Electric Compact Laundry Machine for Drying Clothes
4	13.013	review	B07YWK935R	Perfect fot Amana dryer	rating=5.0, verified=True	Back to drying my clothes, i like
5	12.700	metadata	B08M5XJQYV	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Dryers	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying

Semantic results for: 'small machine for drying clothes in an apartment'

Rank	Score	Source	ASIN	Title	Info	
1	0.491	metadata	B08M5XJQYV	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Dryers	Xiaqing Cl Fast 1000V Machine,P Apartment Electric Clo
2	0.577	review	B08867TL36	This thing is a good little dryer	rating=5.0, verified=True	It does a g clothes at as a load c
3	0.593	metadata	B0BGR136BM	Portable Dryer for Apartments, 800W Portable Dryer for Clothes Mini Dryer Machine for Travel Home Laundry	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Dryers	Portable D 800W Port Mini Dryer Home Lau
4	0.599	metadata	B08KQ2T7HF	Portable Stacked Dryer, Multifunctional Mini Manual Dryer Machine Hand-Operated Dryer Spinner, Non-Electric Compact Laundry Machine for Drying Clothes	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	Portable S Multifunct Dryer Mac Dryer Spin Compact L Drying Clo
5	0.636	metadata	B0C2K7C29Z	Portable Washing Machine,Mini Washer Suitable for Washing Small Pieces of Clothing, Baby Clothes,Underwear,Socks,Portable Washer Machine for Apartments, Dormitories, Camping,RV and More Green	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	Portable W Washer Su Small Piec Clothes,Ur Washer M Dormitorie More Gree

BM25 results for: 'best dishwasher for a small apartment under $800'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	14.233	metadata	B09N8NXR35	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine
2	12.143	metadata	B09MHZ9TBB	VENTRAY Countertop Portable Dishwasher Mini Compact with 5 Washing Programs LED Digital Display for Small Apartment Dorms RVs DW50	Appliances | Appliances > Dishwashers > Countertop Dishwashers	VENTRAY Countertop Portable Dishwasher Mini Compact with 5 Washing Programs LED Digital Display for Small Apartment Dorms RVs DW50
3	10.270	review	B08YNL2SL4	Limited	rating=3.0, verified=False	This portable dryer is limited but useful in a small place like an rv or studio apartment. Assembly was okay and it is easy to use. The wheels are helpful and it is lightweight. However it is...
4	10.246	metadata	B06Y2DV16G	SHE878WD5N 24 800 Series Recessed Handle Dishwasher with 6 Cycles 6 Options Flexible Third Rack Glide Touch Controls in Stainless Steel	Appliances | Appliances > Dishwashers	SHE878WD5N 24 800 Series Recessed Handle Dishwasher with 6 Cycles 6 Options Flexible Third Rack Glide Touch Controls in Stainless Steel
5	9.993	metadata	B07BWLFYVB	Vent Kit, Watts Series 800 M4, 1 in	Industrial & Scientific | Appliances > Parts & Accessories >	Vent Kit, Watts Series 800 M4, 1 in

Rank	Score	Source	ASIN	Title	Info	Snippet
					Dryer Parts & Accessories > Vents	

Semantic results for: 'best dishwasher for a small apartment under $800'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.657	review	B00FKIHUB4	Much cheaper than replacing dishwasher.	rating=5.0, verified=True	Exactly what I expected.
2	0.670	review	B00WAN3R8C	Great dishwasher for the size and price	rating=5.0, verified=True	Great dishwasher for the size and price. We have a tiny kitchen and needed something that fit the space. My husband was able to put this in fairly easily. There is no drying option, but for the...
3	0.692	metadata	B01DSIGF18	Manual Dishwasher, Portable Dishwasher for camping and outdoor	Appliances | Appliances > Dishwashers > Countertop Dishwashers	Manual Dishwasher, Portable Dishwasher for camping and outdoor
4	0.712	review	B00WAN3R8C	Perfect for small kitchens	rating=5.0, verified=True	So glad I could find a small dishwasher for my kitchen. I hate doing dishes, and this thing rocks. I had a contractor install.
5	0.719	metadata	B09MHZ9TBB	VENTRAY Countertop Portable Dishwasher Mini Compact with 5 Washing Programs LED Digital Display for Small Apartment Dorms RVs DW50	Appliances | Appliances > Dishwashers > Countertop Dishwashers	VENTRAY Countertop Portable Dishwasher Mini Compact with 5 Washing Programs LED Digital Display for Small Apartment Dorms RVs DW50

BM25 results for: 'energy-efficient refrigerator for a family of four'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	9.286	review	B00E37TQV0	Perfect Fit	rating=5.0, verified=True	This water filter fit perfect in my refrigerator, would recommend to friends and family
2	8.344	metadata	B0B47KJH4D	Refrigerator Lock Combination, Fridge Lock Combo - Take Care of Your Family with Strongholden - No Keys Needed - Just Stick It (White & Round)	Tools & Home Improvement | Appliances > Refrigerators, Freezers & Ice Makers > Refrigerators	Refrigerator Lock Combination, Fridge Lock Combo - Take Care of Your Family with Strongholden - No Keys Needed - Just Stick It (White & Round)
3	8.177	review	B071J2LSQS	Cheap way to make ice for the family	rating=5.0, verified=True	Si far great and cheap product comparing from all the ones I had in mind to buy for my family.
4	8.153	review	B078MMXJPC	Family Doesn't Understand...	rating=4.0, verified=True	I got this as every time I do dishes and clean the sink out, I thought it would be easier for my family to see whats up with the dishwasher. I keep it on Dirty until its full then start it and...
5	7.401	metadata	B0B3RG3ZPN	Listenwind Matching Christmas Onesies for Family, Family Matching Christmas Onesies Sets Xmas Jumpsuit for Women Men Couples Kids (Women, Small, 001# Red snowmen)	Amazon Home | Appliances > Refrigerators, Freezers & Ice Makers > Beverage Refrigerators	Listenwind Matching Christmas Onesies for Family, Family Matching Christmas Onesies Sets Xmas Jumpsuit for Women Men Couples Kids (Women, Small, 001# Red snowmen)

Semantic results for: 'energy-efficient refrigerator for a family of four'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.723	review	B000SDMQOW	Good purchase	rating=5.0, verified=False	After several months I can say this fridge is awesome. I ordered it sight unseen from a big box store that left me alone to get rid of my old fridge. The local power company hauled away the old...
2	0.743	metadata	B0B6Z36SC1	KRIB BLING 3.5 Cu.ft Retro Mini Fridge with Freezer - Compact Refrigerator for Home, Office, Dorm, or RV with Adjustable Mechanical Thermostat and 2-Door Design, Wood	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Refrigerators	KRIB BLING 3.5 Cu.ft Retro Mini Fridge with Freezer - Compact Refrigerator for Home, Office, Dorm, or RV with Adjustable Mechanical Thermostat and 2-Door Design, Wood
3	0.757	metadata	B093BPW8WY	FRIGIDAIRE Portable 10L, 15-can Mini Fridge Brushed Stainless Rugged Refrigerator, EFMIS183-SS (Renewed)	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Refrigerators	FRIGIDAIRE Portable 10L, 15-can Mini Fridge Brushed Stainless Rugged Refrigerator, EFMIS183-SS (Renewed)
4	0.770	review	B00ZYE4CNI	Inexpensive and have a fourpack	rating=4.0, verified=True	The delivery was quick and I have installed one of the four. The fridge uses only one filter but I bought the 4 pac pack because it was available. The filter is difficult for me to reach, mounted...
5	0.801	review	B00O2CF24Q	Bang for the buck!	rating=5.0, verified=True	Great for the price, keeps fridge from smelling!

BM25 results for: 'reliable stove for frequent home cooking that is easy to clean'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	14.897	metadata	B08YDWSKP6	Stove Burner Covers, 4 Pack, Gas Stove Burner Liners, Non-stick Reusable Gas Range Stove Protector Covers for Kitchen, Cooking, Cuttable, Easy to Clean JORETLE(Black)	Tools & Home Improvement | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	Stove Burner Covers, 4 Pack, Gas Stove Burner Liners, Non-stick Reusable Gas Range Stove Protector Covers for Kitchen, Cooking, Cuttable, Easy to Clean JORETLE(Black)
2	14.375	review	B07M8JWVN1	Easy to clean stove.	rating=4.0, verified=True	Easy to clean your stove. The only thing is you need to cut them as per your stove size.
3	12.953	review	B073PLT7TY	Not easy to clean	rating=5.0, verified=True	For my stove fine. Not easy to clean. Already need new ones.
4	12.543	review	B007EEG3AG	A good fit	rating=5.0, verified=True	I like the way these stove cover fit (some do not fit well). This helps to keep the clean up after cooking to barely nothing,
5	12.032	review	B092RXSMGD	Easy to use	rating=5.0, verified=True	Loved how easy it was to put on and helps to keep the stove clean

Semantic results for: 'reliable stove for frequent home cooking that is easy to clean'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.691	review	B073PLT7TY	Not easy to clean	rating=5.0, verified=True	For my stove fine. Not easy to clean. Already need new ones.
2	0.794	review	B0763BMN9T	A must have for your stove	rating=5.0, verified=True	These make cleaning the stove so easy. Recommend to all.
3	0.811	review	B091TK8WYG	Great price, work great!	rating=5.0, verified=False	The price on these is great! I have a 4 burner stove so I like having a spare set to swap out with when the other needs cleaning. I throw it in the top rack of the dishwasher and it cleans them...
4	0.814	review	B00UNOSLYU	Great buy	rating=5.0, verified=True	I really like this cooker, It gets hot very quickly and cooks evenly. It's also very easy to clean up afterwards. Great alternative if you don't have a stovetop available. You just have to buy the...
5	0.818	review	B0037Z7HQK	Great Cooktop	rating=5.0, verified=True	Just moved from my beautifully remodeled kitchen with a gas range to an older home with a coiled electric cooktop. This portable Induction burner is now my sole cooktop. I love it. The ease of...